Lab 09

ENVX2001 Applied Statistical Methods

Contents

Welcome to week 9 - the last week of modelling!

This week's lab will give you some practice testing how well our models predict new values.

Next week you will start on multivariate approaches.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use or the statistical programming language R to apply an ANOVA and fit regression models to experimental data
- LO5. interpret the output and understand conceptually how its derived of a regression, ANOVA and multivariate analysis that have been calculated by R
- LO6. write statistical and modelling results as part of a scientific report

In this lab, we will learn how to:

1. Use the parsimonious models we produced last week to predict new values
2. Interpret key prediction quality metrics - do our models predict well?

Specific goals

By the end of this lab, you should be able to:

- ☐ Subset data into training and test sets
- ☐ Fit models to the training set and make predictions
- ☐ Plot predicted vs observed values to understand the agreement between the two
- ☐ Calculate and interpret confidence and prediction intervals
- ☐ Calculate and interpret error metrics (mean error, MAE, RMSE)
- ☐ Calculate linearity metrics r^2 and Lin's correlation coefficient

Preparation

Please work on these exercises by creating your own Quarto file. You are welcome to write your responses up in the style of a report.

Packages

This lab uses `tidyverse`, `readxl`, `moments`, `psych`, `performance`, `caret`, `epiR`. Install any you are missing by running the following **in the console**:

```
CODE
install.packages(c("epiR", "caret", "tidyverse", "readxl", "moments", "psych", "performance"))
```

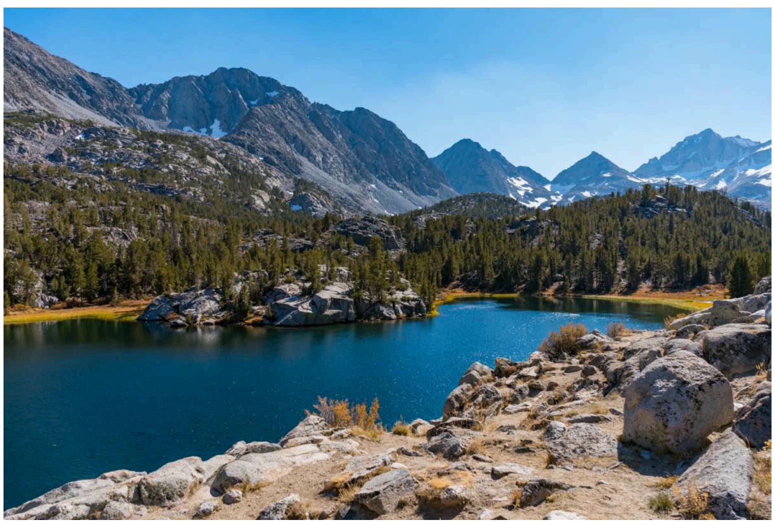
If a package is already installed, R will simply reinstall it — no harm done.

Downloads

File	Used in	Download
`california_streamflow.xlsx`	Section 1	Download
`plant_aridity.csv`	Section 2	Download

Save both files into a folder called `data` inside your project folder.

1. Making predictions with California streamflow (~45 min)



Last week we found that precipitation at Lake Sabrina was not a significant predictor of runoff volume near Bishop, California.

Our Partial F-test between our full and reduced model resulted in a P-value< 0.05, so we failed to reject the null hypothesis. This told us that there was no significant difference between full and reduced models, so we selected the reduced model as it was less complex and therefore more parsimonious.

Our selected model was:

```
CODE
reduced_model <- lm(runoff_volume ~ pine_creek + rock_creek, data = stream_data)
```

This week we will use this optimal model to predict new values of runoff volume near Bishop, California.

We will then assess the prediction quality by comparing observed to predicted values and calculating validation metrics to assess how well this model predicts new values.

Data recap

The california_streamflow dataset you were analysing contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. You will focus on only 3 variables.

The data is in the **california_streamflow.xlsx** file.

Variables are as follows:

- `lake_sabrina`: Precipitation recorded at Lake Sabrina (mm)
- `pine_creek`: Precipitation recorded at Big Pine Creek (mm),
- `rock_creek`: Precipitation recorded at Rock Creek (mm)
- `runoff_volume`: Stream runoff volume measured at a site near Bishop, California (ML/year). This will be the response variable.

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

Data source: Hollett, K.J., Danskin, F.M., McCaffrey, W.F., and Walti, S.L., 1991. Geology and water resources of Owens Valley, California. U.S. Geological Survey Water-Supply Paper 2370-H.

Exercise 1 - Making predictions

(i) Read the data in. What is the sample size? How do you think the sample size might impact the prediction quality?

(ii) Split the data into training and test subsets. We recommend an 80/20 training/test split.

(iii) Check the sample size and distribution of each subset. Does the sample size of each subset reflect the 80/20 split? Are the distributions (mean, median, min, max) similar between the two subsets?

(iv) Fit the linear model to the training subset using runoff_volume as the response, and pine_creek and rock_creek as the predictor variables.

(v) Check whether assumptions have been met for your trained model. Have the assumptions been met? or do we need to explore transformation?

(vi) Use the model to predict using the training set and then using the test set.

(vii) Have a look at what the `predict()` function has produced using the `head()` function. We should see a new column at the end of both dataframes containing the predictions from the model.

Exercise 2 - How well does our model predict new values?

(i) Plot observed vs predicted for the training and test subsets. Do the points follow the 1:1 line? Do the points sit close to the line, or are they scattered about the plot area?

(ii) Overlay the confidence interval onto the training predicted vs observed plot. What does this tell us?

(iii) Overlay the prediction interval onto the predicted vs observed plot produced using the test subset. What does this tell us?

(iv) Calculate the mean error for predictions obtained with the training and test subsets. Are they similar to each other? Is the Mean Error close to 0?

(v) Calculate the Mean Absolute Error. Are the values close to zero? what does this tell us?

(vi) Calculate the Root Mean Square Error. Are the values close to zero? what additional information can this tell us about the accuracy of our predictions?

HINT: you can use the `RMSE()` function from the `caret` package

(vii) Calculate the r^2 value. What does this metric tell us about our model? Is the r^2 good for our predictions?

HINT: you can either calculate this by squaring the correlation coefficient between observed and predicted, or you can use the `R2()` function from the `caret` package.

(viii) Calculate Lin's Correlation Coefficient for our predictions. What does this metric tell us about our model? How is this different from our r^2 value? Do our values indicate strong agreement between observed and predicted for our training and test predictions?

HINT: you can calculate the LCCC via the `epiR` package

(ix) Create a summary table of the metrics above and use it to make a conclusion about the prediction quality of our model.

Before we move on, now is a good time to take a 5-minute break.

2. Making predictions with the plant aridity data (~45 min)

Figure 1: Image of invasive capeweed. Source: Adobe Stock.

We used this dataset last week as an example of model selection.

As identified by the partial F-test and stepwise regression last week, we found that the reduced model was most appropriate, which includes `growth rate` and `chlorogenic_acid` as the predictors. This reduction in predictors made our model more significant, though it only explained 7% of variation in log(rel_fitness).

Although the model does not explain much variation, we will use this data to get some practice making predictions, and to see what happens when we use a sub-optimal model.

Data recap

This dataset examines the adaptation of invasive capeweed (*Arctotheca calendula*) populations to aridity gradients across southern Australia. A common garden experiment was conducted from 2018-2019 with populations sourced from varying aridity levels, subjecting them to wet and dry treatments to measure traits related to drought escape, avoidance, and tolerance.

The dataset has been reduced to contain only the variables you will need to fit in the model. Furthermore, only the samples that received the wet treatment will be used for this analysis (n = 89 observations).

The dataset is in the *plant_aridity.csv* file.

Variables are as follows:

- `growth_rate`: growth rate measured as differences in # of rosette leaves between zero and eight weeks after transplanting
- `days_to_flower`: days to first flowering since transplant (days)
- `specific_leaf_area`: specific leaf area as [leaf area (cm2) / dry mass (g)]
- `root_shoot_ratio`: root-to-shoot ratio [root mass/shoot mass]
- `chlorogenic_acid`: concentration of chlorogenic acid (mg/g frwt sample)
- `rel_fitness`: Relative fitness of each plant, seed count divided by the mean seed count

`rel_fitness` will be the response variable in your models, and the other variables will be the predictor variables.

Data source: Uesugi, Akane (2022). Data from: Multivariate selection mediated by aridity predicts divergence of drought resistant traits along natural aridity gradients of an invasive weed [Dataset]. Dryad. https://doi.org/10.5061/dryad.tht76hf17

Paper: Carvalho C, Davis R, Connallon T, Gleadow RM, Moore JL, Uesugi A. Multivariate selection mediated by aridity predicts divergence of drought-resistant traits along natural aridity gradients of an invasive weed. New Phytol. 2022 May;234(3):1088-1100. doi: 10.1111/nph.18018.

Exercise 3 - Making predictions

(i) Read the data in. What is the sample size? How do you think the sample size might impact the prediction quality?

(ii) Before we do any splitting, make sure you log-transform `relative_fitness` by creating a new variable within the plant_aridity dataframe, as this is what we found was important in our model last week.

(iii) Split the data into training and test subsets. We recommend an 80/20 training/test split.

(iv) Check the sample size and distribution of each subset. Does the sample size of each subset reflect the 80/20 split? Are the distributions (mean, median, min, max) similar between the two subsets?

(v) Fit the linear model to the training subset using log(rel_fitness) as the response, and growth_rate and chlorogenic_acid as the predictor variables.

(vi) Check whether assumptions have been met for your trained model. Have the assumptions been met? or do we need to explore transformation?

(vii) Use the model to predict using the training set and then using the test set.

(viii) Have a look at what the `predict()` function has produced using the `head()` function. We should see a new column at the end of both dataframes containing the predictions from the model.

Exercise 4 - How well does our model predict new values?

(i) Before we continue, we will need to backtransform our predicted and observed values for the training and test subsets by exponentiating them.

(ii) Plot observed vs predicted for the training and test subsets. Do the points follow the 1:1 line? Do the points sit close to the line, or are they scattered about the plot area?

(iii) Calculate the confidence interval and backtransform it. Overlay the backtransformed confidence interval onto the training predicted vs observed plot. What does this tell us?

(iv) Calculate the prediction interval and ensure you backtransform it. Overlay the prediction interval onto the predicted vs observed plot produced using the test subset. What does this tell us?

(v) Calculate the mean error for predictions obtained with the training and test subsets. Are they similar to each other? Is the Mean Error close to 0?

(vi) Calculate the Mean Absolute Error. Are the values close to zero? what does this tell us?

(vii) Calculate the Root Mean Square Error. Are the values close to zero? what additional information can this tell us about the accuracy of our predictions?

HINT: you can use the `RMSE()` function from the `caret` package

(viii) Calculate the r^2 value. What does this metric tell us about our model? Is the r^2 good for our predictions?

HINT: you can either calculate this by squaring the correlation coefficient between observed and predicted, or you can use the `R2()` function from the `caret` package.

(ix) Calculate Lin's Correlation Coefficient for our predictions. Do our values indicate strong agreement between observed and predicted for our training and test predictions?

HINT: you can calculate the LCCC via the `epiR` package

(x) Create a summary table of the metrics above and use it to make a conclusion about the prediction quality of our model.

Conclusion

Closing thoughts

You have done a great job investigating how well our chosen models predict new values of the response.

We saw a good example of a model that predicts new values well with the california streamflow data, and a good example of a model that does not predict new values well with the plant aridity data.

We have also seen the importance of visualisation and metrics to assist us in understanding the prediction quality of our models. There is no single metric that will give us the magic answer; it is up to us to make an informed decision on how well our model can predict.

Thank you for all of your hard work and participation in this module. Keep up the great work and enjoy learning about multivariate approaches from next week!

If you have any questions, please approach your tutors or you are welcome to post to the Ed Discussion board.

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